

You are generating new competency questions for validating and updating the FASTO ontology. This is a non-interactive task. Use the GPT 5.5 High Model. Return exactly one valid downloadable JSON file and nothing else.

Do not:

- ask questions;
- request clarification;
- summarize the ontology;
- explain your reasoning outside the JSON.

Task

Generate 15-20 substantively new competency questions that assess FASTO's current coverage and identify evidence-supported ontology update needs. Do not generate questions solely to make FASTO fail.

Evidence

Six evidence sources are provided.

1. Primary evidence: The original FASTO_RDF.owl in RDF/XML syntax. This is the authoritative source.
2. Secondary evidence: A rule-based verbalizer generated from a temporary sanitized copy of FASTO. This exists only to improve readability.
3. FASTO paper: Use it to clarify FASTO's scope, modeled concepts, and design goals.
4. Original competency questions: The competency questions from the FASTO paper. Use them only as an exclusion set to prevent duplication.
5. Official sources: Current versions of the official sources (12) used by the FASTO paper. Use them to identify the latest standards, terminology, and recommendations for Type 1 diabetes management and treatment that may support new competency questions and ontology update needs.

Pay particular attention to:

- care for children;
 - age-related differences in management and treatment;
 - Treatment of hypoglycemia.
6. Current domain literature: Current papers (16) on Type 1 diabetes management and treatment. Use them to identify relevant validation topics and ontology update needs.

Evidence rules

- Treat the FASTO ontology as authoritative for existing ontology content.
- Use the verbalizer only to improve readability.
- Use the paper to clarify FASTO's intended scope and coverage.
- Use the current official sources to identify evidence-supported changes to motivate new competency questions or update needs. FASTO was released in 2018.
- Generate questions only when their relevance is supported by the supplied evidence.
- Do not invent medical facts, ontology entities, relations, sources, or intended semantics.
- Cite the evidence motivating each question.
- If external research is used, identify the source in `additional_comment`.
- Use lower confidence when evidence or coverage is uncertain.
- Do not treat an empty SPARQL result as proof that the ontology lacks coverage.

Instructions

- Do not repeat, paraphrase, narrow, combine, or slightly reformulate an original competency question.
- A generated question must test a substantively different capability, relation, constraint, temporal aspect, classification, or update need.
- Avoid overlap among the newly generated questions.
- Identify existing FASTO entities required for each question.
- Suggest new entities only when the required concepts or relations are absent.
- Distinguish ontology modeling gaps from missing instance data.
- Translate each question into SPARQL when possible.
- Do not invent ontology entities in SPARQL queries.
- If SPARQL translation is partial or impossible, explain the limitation.
- Report any resulting gap or update need supported by evidence.

Reporting rules

- Return exactly one valid JSON object.
- Generate between 15 and 20 competency questions.
- Include all required fields and arrays, even when empty.
- Use sequential IDs: CQ1, CQ2, CQ3, and so on.
- Use confidence values from 0.0 to 1.0.
- Use valid JSON strings and escape quoted evidence correctly.

Field rules

expected_answer_type: One of:

- boolean
- entity_list
- value
- count
- relation_path
- classification
- temporal
- other

sparql_translatable: One of:

- yes
- partly
- no

ontology_coverage: One of:

- covered
- partly_covered
- not_covered
- unclear

coverage_gap_type: One of:

- none
- missing_entity
- missing_relation
- missing_axiom
- insufficient_modeling
- requires_instance_data
- outside_scope
- unclear

priority: One of:

- high
- medium
- low

Required JSON schema

```
{
  "task": "new_competency_question_generation_and_coverage_analysis",
  "model": "",
  "input_condition": "llm_only",
  "ontology": {
    "name": "FASTO",
    "scope_summary": ""
  },
  "competency_questions": [
    {
      "id": "CQ1",
      "question": "",
      "motivation": "",
      "evidence": [],
      "related_topic": "",
      "expected_answer_type": "other",
      "required_existing_entities": [],
      "suggested_new_entities": [],
      "sparql_translatable": "no",
      "sparql_query": "",
      "sparql_limitation": "",
      "ontology_coverage": "unclear",
      "coverage_evidence": "",
      "coverage_gap_type": "unclear",
      "related_gap_or_update_need": "",
      "priority": "medium",
      "confidence": 0.0,
      "additional_comment": ""
    }
  ],
  "coverage_summary": {
    "number_of_questions": 0,
    "number_covered": 0,
    "number_partly_covered": 0,
    "number_not_covered": 0,
    "number_unclear": 0,
    "covered_topics": [],
    "missing_or_underrepresented_topics": [],
    "main_update_needs": [],
    "comments": ""
  }
}
```

Input

Official sources:

- <http://www.open-mhealth.org>
- <http://www.opencem.org>
- www.nice.org.uk
- www.guideline.gov
- www.diabetes.org
- www.diabetes.ca
- www.medscape.com
- www.drugs.com
- www.medicinenet.com
- www.uptodate.com
- www.fda.gov
- <http://sideeffects.embl.de/>

Internal quality checks

Before returning the JSON, internally verify that:

- every generated question is substantively different from all original competency questions and from the other generated questions;
- every question is supported by evidence;
- required existing entities occur in FASTO;
- suggested entities are clearly distinguished from existing entities;
- SPARQL queries use only supplied ontology entities;
- coverage assessments distinguish ontology gaps from missing instance data;
- the coverage summary counts match the question assessments;
- the final output is syntactically valid JSON.

Now analyze the supplied ontology evidence and return the downloadable JSON file only.